9. Compute

$$\int \frac{1}{1+x^2}\, dx.$$

10. Compute

$$\int x\ln(x)dx \quad \text{and} \quad \int xe^x dx.$$

11. Compute

$$\int x^n \ln(x)\, dx.$$

12. Compute

$$\int (\ln(x))^n\, dx.$$

13. Solve the ODE

$$y'' - 4y' + 4y = 1.$$

14. Find $f(x)$ such that

$$f'(x) \;=\; f(x)(1 - f(x)).$$

15. Derive the Black-Scholes PDE.